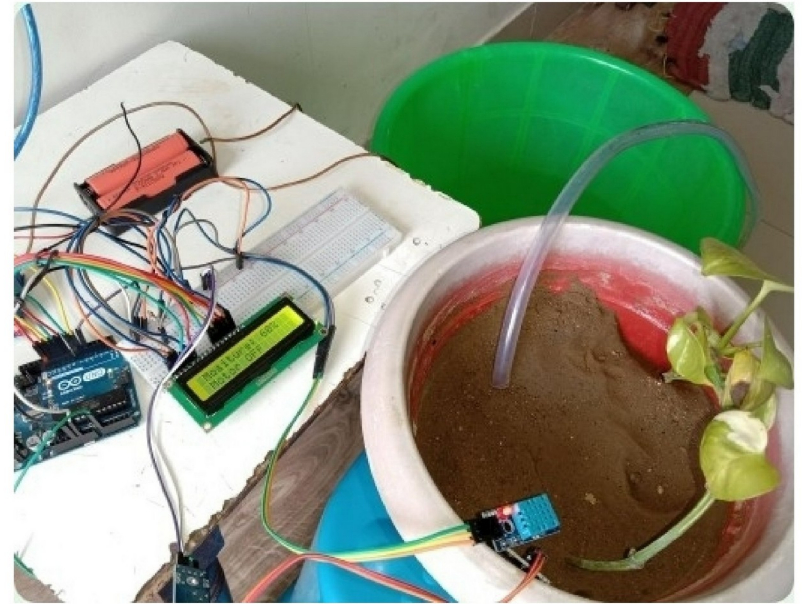


Smart Irrigation System

I built a **Smart Irrigation System** that manages water for plants automatically.

- It measures moisture level in the soil.
- It controls the motor pump without human help.
- It uploads live weather and soil data to the cloud.



Why we need this?



Under-Watering

When people forget to water them, plants dry up and die.



Over-Watering

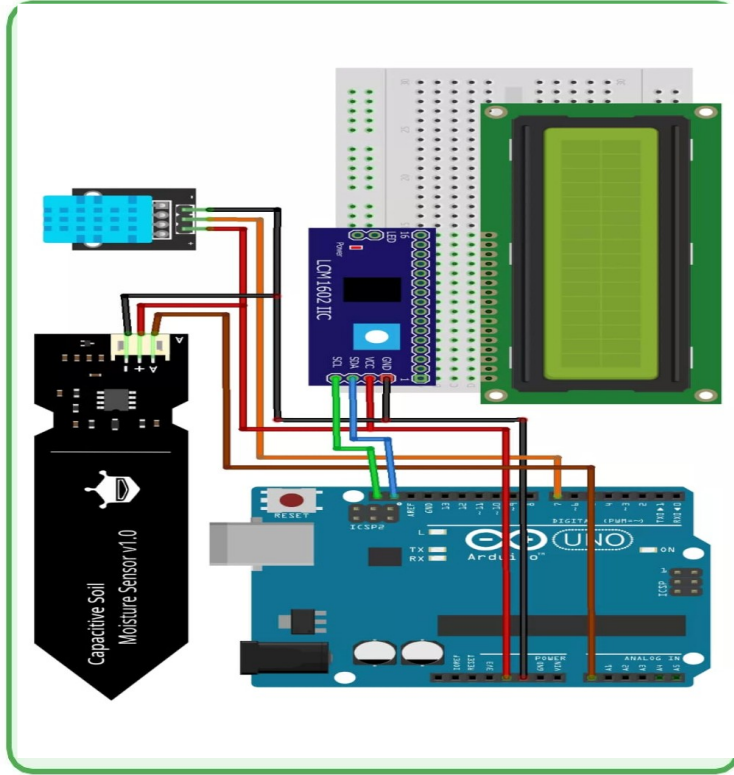
Giving too much water wastes resources and damages plant roots.



Manual Effort

Busy lifestyles make it hard to check plants daily.

Main Hardware Components



Arduino Uno: Main brain – controls everything



NodeMCU (ESP8266): Adds Wi-Fi for internet connection



Soil Moisture Sensor: Reads how wet the soil is



DHT11 Sensor: Measures temperature and humidity



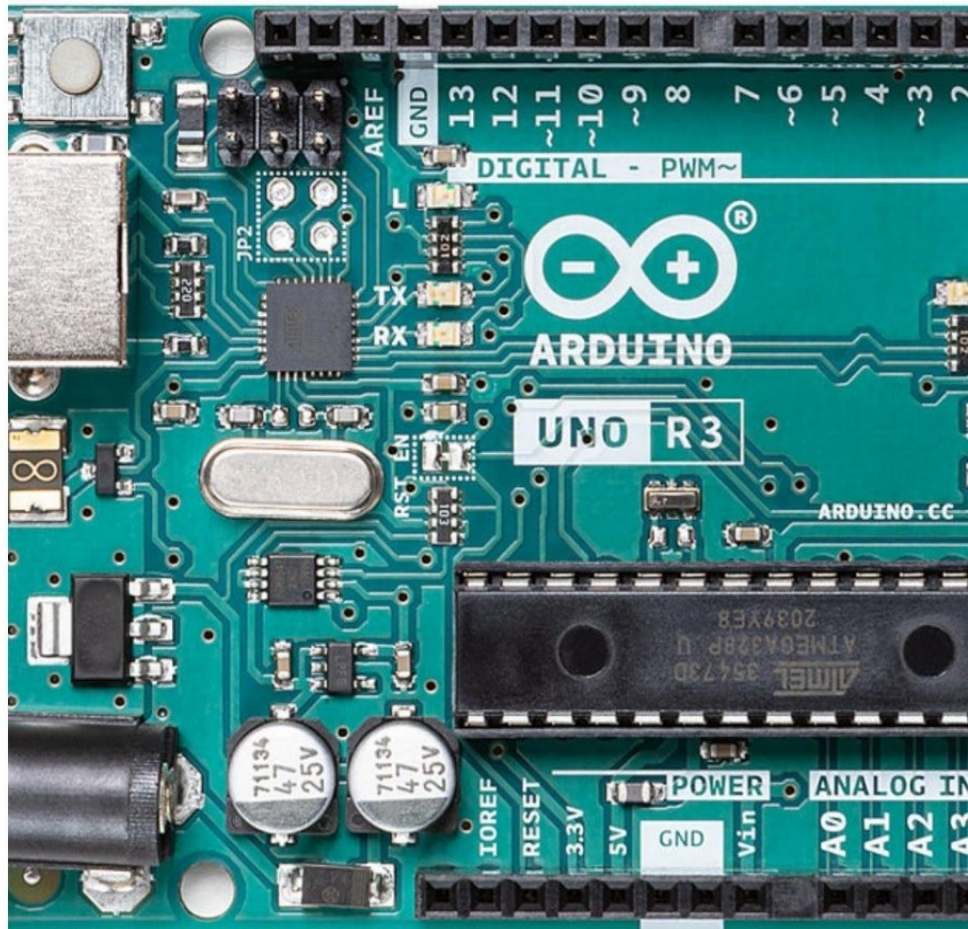
5V Relay & Motor Pump: Controls water flow on/off

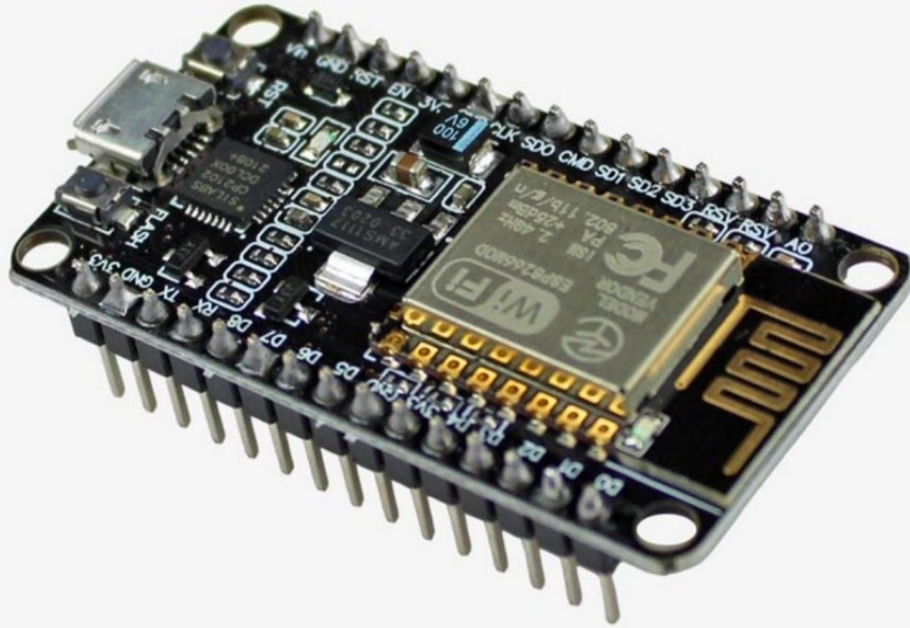
ARDUINO UNO

The **Brain** of our project. It reads all sensor data and makes smart decisions instantly.

```
void loop() {  
  int val = analogRead(A1);  
  if(val < 30) pumpOn();  
}
```

It connects all hardware parts together.





NODEMCU ESP8266

Our **Cloud Link**. It adds Wi-Fi features so the project can talk to the internet.

- Sends live stats to the Blynk App.
- Allows remote control via smartphone.
- Syncs soil and weather data every 2 seconds.



ULTRASONIC SENSOR

Used to measure the **Water Tank Level**. It uses sound waves to check depth.

- HC-SR04 uses high-frequency sound pulses.
- Tells us if the tank is empty or full.
- Protects the pump from running dry.

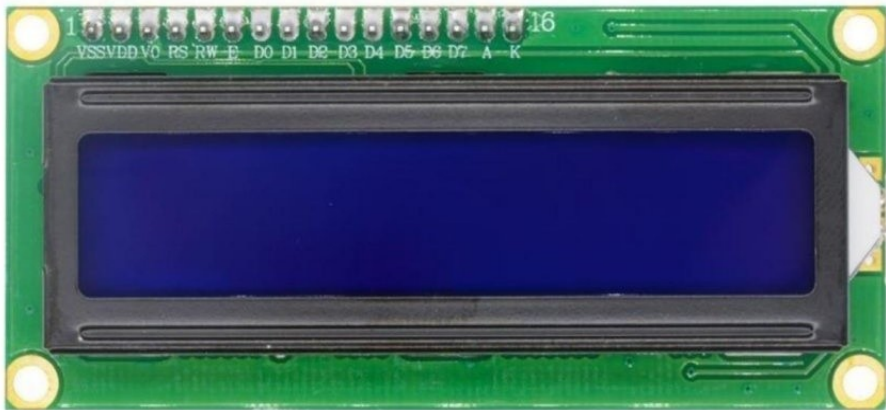


WATER PUMP / MOTOR

The **Hand** of the project. It physically moves the water to the plants.

```
digitalWrite(7, LOW); // Start Motor  
delay(5000); // Water for 5 sec  
digitalWrite(7, HIGH); // Stop
```

Efficient flow with low power consumption.



16X2 I2C LCD

The **Status Screen**. It shows moisture %, tank level, and temperature live.

```
lcd.setCursor(0,0);  
lcd.print("Soil: "); lcd.print(percent);
```

Allows local monitoring without any screen.



| PIEZO BUZZER

The **Alarm** speaker. It makes sounds when something is wrong.

- Beeps if the water tank level is too low.
- Sounds an alert if a sensor fails to read.
- Warns the user during high heat alerts.

JUMPER WIRES

The **Veins** of the project. They carry electricity and data signals.

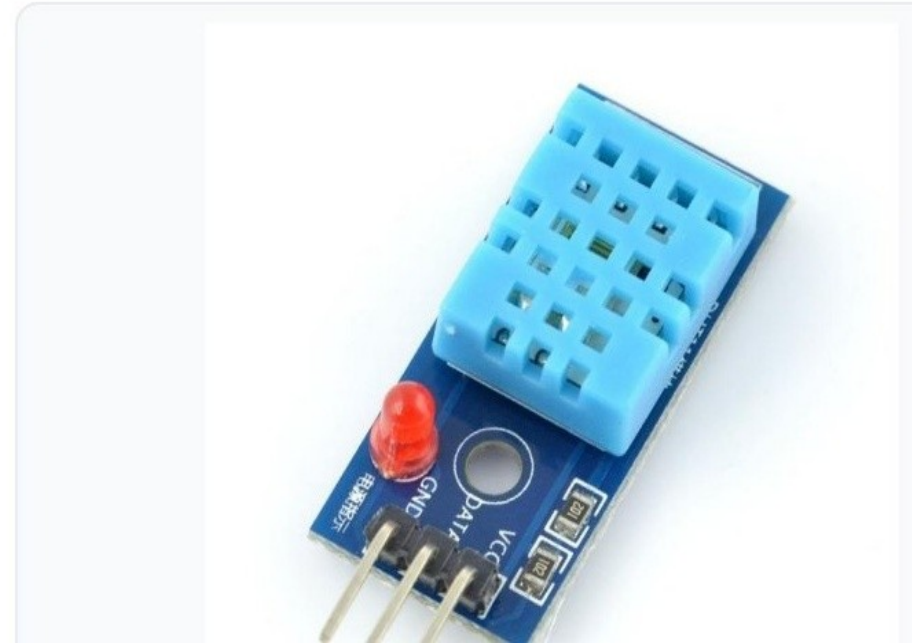
- **Male-to-Male:** For breadboard.
- **Male-to-Female:** To connect sensors.
- Colorful wires for easy troubleshooting.



| DHT11 CLIMATE SENSOR

Tracks **Temperature** and **Humidity**. Helps us know the weather around the plant.

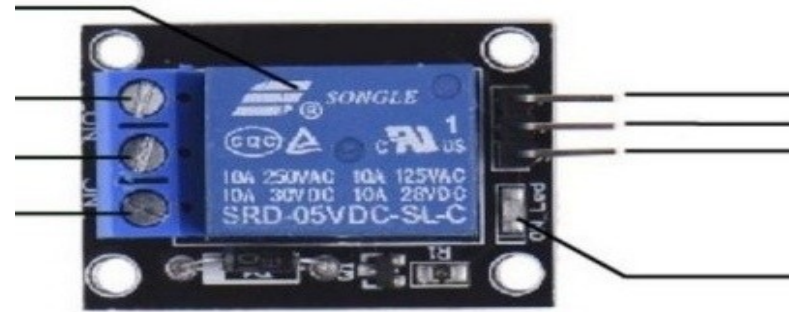
- Measures temperature: 0°C to 50°C
- Measures humidity: 20% to 90%
- Sends data to Arduino every 2 seconds
- Protects plants from heat stress



| 5V RELAY MODULE

The **Switch**. Since Arduino cannot power a motor directly, the relay acts as an electronic switch.

- Controls the water pump ON/OFF
- Triggered by Arduino digital signal
- Handles high voltage loads safely
- Code: `digitalWrite(7, LOW); // Trigger Relay`



WORKING MECHANISM



Step 1: Sensors

Sensors check the soil moisture and tank water level every second.



Step 2: Logic

Arduino checks if soil is Dry (<30%). If yes, it checks the tank level.



Step 3: Output

If tank has water, the Relay turns ON and the Pump waters the plant.

Data is synced to the Blynk App simultaneously!

KEY ADVANTAGES



Healthy Plants

Plants never die from drying or over-watering. They stay green!



Save Water

The system only uses the exact water needed. No wasting water.



Set & Forget

The machine works day and night. You don't have to check it daily.

Technology making nature safer and gardening easier!

| Applications

- **Home Gardens:** Waters your plants automatically when you are not at home or busy.
- **Greenhouses:** Controls water and monitors heat for sensitive or expensive plants.
- **Smart Farms:** Helps farmers save water and grow better crops in large fields.
- **Roof Gardens:** Perfect for small vertical gardens on rooftops in modern cities.



| Drawbacks

- **Needs Power:** The system stops working if the electricity or battery dies.
- **Needs Power:** The system stops working if the electricity or battery dies.
- **Sensor Rust:** The soil sensor probe can get damaged or rust inside wet mud quickly
- **Roof Gardens:** Perfect for small vertical gardens on rooftops in modern cities.

